

Robust Individual and Group Fair Classification Under Adversarial Data Corruption

Implementation and Empirical Evaluation of DRO-FAIR (Algorithm 1, ICML Submission)

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Abstract. We implement and evaluate **DRO-FAIR**, a distributionally robust optimization approach for joint Demographic Parity (DP) and Individual Fairness (IF) under data corruption, following the exact Algorithm 1 from the ICML submission. Our key contribution replaces the paper’s random noise with **multi-modal adversarial corruption** — PGD-based feature attacks, coordinated label flips, and minority-targeted attribute flips — providing a 2–5 $\times$ harder evaluation at the same corruption level α . On the complete canonical grid (540 rows; fixed $\tau=1$, $k_{\text{inner}}=10$, $n=6$ seeds), DRO-FAIR achieves significantly lower DP than Naive-FAIR on **Adult** and **Credit** at $\alpha \leq 0.2$ under the DP-targeted and Combined attacks (Wilcoxon $p<0.05$; Adult/DP $\alpha=0.1$ is 5/6, others 6/6 in that regime), reproducing the original Jun 30 report for Adult DP, with accuracy equal-or-better. The previous “DRO fragile” failure on Adult was a high-temperature ($\tau=100$) artifact. LSAC/DP is a degenerate negative result (constant-predictor collapse, DRO loses at 0/6 seeds); LSAC/Combined is a genuine win ($p=0.0156$). IF-attack results are complete and mixed on DP (180 rows): IF metric wins Adult/Credit at $\alpha \in \{0.1\text{--}0.4\}$ incl. $\alpha=0.3$ (6/6, $p=0.0156$); DP-under-IF remains mixed (Adult $\alpha=0.3$ DP loss; LSAC/IF DP loss) — see `results/if_wilcoxon_summary.txt`. Pre-fix IF metrics were degenerate ($\approx 10^{-10}$). On real UTKFace ResNet18 features (complete 90-config REAL grid) clean-test DP is also mixed (significant DRO wins mainly at high α ; not an Adult/Credit $\alpha \leq 0.2$ copy). All theoretical formulas are numerically verified. Code, results, and diagnostics are fully reproducible (`docs/KULDEEP_CORRECTION.md`, `paper/sections/results.tex`).

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1. Introduction

Fairness-aware classifiers trained on corrupted data can appear fair on training data while violating fairness on the true distribution. DRO-FAIR [Anonymous, 2026] addresses this by solving a min-max Lagrangian where the inner maximisation searches for the worst-case reweighting within TV uncertainty sets calibrated to corruption level α .

Our contribution over the paper

Table 1: Adversarial vs. random corruption components.

Attack Component	Random (Paper)	Adversarial (Ours)
Feature perturbation	Gaussian noise $\mathcal{N}(0, \varepsilon^2)$	PGD: $x' = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L})$, $\ \delta\ _\infty \leq 0.1$
Label flips	Uniform random	Coordinated to maximise DP gap
Attribute flips	Uniform random	70% targeted at minority group
Effect at $\alpha = 0.2$	DP increase $\approx +0.01$	DP increase $\approx +0.03\text{--}0.05$ (3–5 $\times$)

2. Problem Formulation

2.1 Setup and Notation

Let $\mathcal{D} = \{(x_i, y_i, a_i)\}_{i=1}^n$ with $x_i \in \mathbb{R}^d$, $y_i \in \{0, 1\}$, $a_i \in \{0, 1\}$. An α -corruption adversary modifies $\lfloor \alpha n \rfloor$ samples. The classifier uses soft predictions $\tilde{h}_\theta(x) = \sigma(\tau f_\theta(x))$ with temperature τ .

Definition 1 (Demographic Parity). *A classifier satisfies ε_{DP} -DP if $|\mathbb{E}[\tilde{h}(X) \mid A = 1] - \mathbb{E}[\tilde{h}(X) \mid A = 0]| \leq \varepsilon_{\text{DP}}$.*

Definition 2 (Individual Fairness). *A classifier satisfies ε_{IF} -IF (k -NN) if $\frac{1}{n-1} \sum_{(i,j) \in \mathcal{N}_k} (|\tilde{h}(x_i) - \tilde{h}(x_j)| - d(x_i, x_j) - \gamma)_+ \leq \varepsilon_{\text{IF}}$.*

3. Method: DRO-FAIR

3.1 Corruption-Calibrated TV Radii

Theorem 1 (DP Radii — Theorem 4.2 of Anonymous 2026). *For group $j \in \{0, 1\}$ with true population proportion π_j :*

$$\rho_{\text{DP},j} = \frac{\alpha}{(1 - \alpha)\pi_j + \alpha} \quad (1)$$

Theorem 2 (IF Radius — Theorem 4.3 of Anonymous 2026).

$$\rho_{\text{IF}} = 2\alpha - \alpha^2 \quad (2)$$

Bias-corrected proportions (Appendix F). Since training data is corrupted, observed $\hat{\pi}_j$ is biased. The clean estimate is $\pi_j = (\hat{\pi}_j - \alpha)/(1 - 2\alpha)$, clipped to $[0, 1]$. This is applied in `_compute_radii()` of `dro_fair.py`.

Table 2: TV radii for Adult group proportions ($\pi_0 = 0.67$, $\pi_1 = 0.33$). Minority group gets a larger radius — greater proportion uncertainty. ℓ_1 -ball radius = 2ρ (TV $\rightarrow \ell_1$ conversion).

α	$\rho_{\text{DP},0}$ (maj.)	$\rho_{\text{DP},1}$ (min.)	ρ_{IF}	$2\rho_{\text{DP},1}$ (ℓ_1 radius)
0.0	0.0000	0.0000	0.0000	0.0000
0.1	0.1422	0.2519	0.1900	0.5038
0.2	0.2717	0.4310	0.3600	0.8621
0.3	0.3901	0.5650	0.5100	1.1299
0.4	0.4988	0.6689	0.6400	1.3378

3.2 Optimization Objective

$$\min_{\theta} \max_{\lambda \geq 0} \left[L_{\text{tilt}}(\theta) + \lambda_{\text{DP}} \cdot \max_{\tilde{p} \in \mathcal{U}_{\text{DP}}} g_{\text{DP}}(\tilde{h}_\theta, \tilde{p}) + \lambda_{\text{IF}} \cdot \max_{\tilde{p} \in \mathcal{U}_{\text{IF}}} g_{\text{IF}}(\tilde{h}_\theta, \tilde{p}) \right] \quad (3)$$

where $L_{\text{tilt}}(\theta) = \beta \log(\frac{1}{n} \sum_i e^{\ell_i/\beta})$ is the tilted risk ($\beta = 5$, approximates CVaR), g_{DP} is the weighted group rate difference, and g_{IF} is the weighted k -NN violation.

3.3 Algorithm 1 — Three-Step Update (exact paper order)

- Forward pass:** $\tilde{h} = \sigma(\tau \cdot f_\theta(X))$, $\tau = 1$ (fixed).
- Outer minimisation (θ):** AdamW step on $L_{\text{tilt}} + \lambda_{\text{DP}} g_{\text{DP}} + \lambda_{\text{IF}} g_{\text{IF}}$; gradient clip $\|\nabla\| \leq 0.5$.
- Dual ascent (λ):** $\lambda \leftarrow \text{clamp}(\lambda + \eta_\lambda \cdot 0.95^t \cdot g, 0, \lambda_{\text{max}})$ [decaying λ -lr for stability].

4. **Inner maximisation** ($\tilde{p}$, $K = 10$ steps): gradient ascent on $g(\theta, \tilde{p})$ [*not* λg — same argmax, avoids instability] $\rightarrow$ project onto $\Delta_n \cap B_1(\tilde{p}, 2\rho)$ via Dykstra’s alternating projection (max_iter=500).

Table 3: Hyperparameters. All from paper §7.1/§G.4 unless noted.

Parameter	Value	Source	Justification
Epochs	60	§7.1	Paper training budget
K_{inner}	10	§G.4	Inner PGD steps per epoch
η_{θ} (AdamW)	10^{-3}	§G.4	Standard AdamW default
η_{λ} (dual)	5×10^{-3}	§G.4	Faster convergence than θ
η_p (inner max)	5×10^{-3}	§G.4	Matches λ scale
λ_{max}	1.5	Stability	Prevents λ_{DP} runaway on Adult
τ (fixed)	1	§G.6	Soft predictions; critical finding
β (tilted risk)	5.0	§G.5	$\beta \rightarrow \infty$: ERM; $\beta \rightarrow 0$: max; $5 \approx \text{CVaR}$
k (k-NN, IF)	5	§7.1	Neighbourhood for metric fairness
γ (IF slack)	0.0	§7.1	Strict metric fairness
Weight decay	10^{-4}	§G.4	L2 regularisation
Grad clip norm	0.5	Stability	Tight clipping to prevent λ instability
λ LR decay	0.95^t	Stability	Decaying λ -update prevents runaway at high α

4. Experimental Setup

Table 4: Dataset summary. Adult has $8\times$ larger baseline DP than Credit/LSAC, making it the hardest test case for DRO-FAIR.

Dataset	Samples	Features	Protected	Task	Baseline DP
Adult	45,222	12	Sex (binary)	Income >\$50K	≈ 0.17 (large)
Credit	30,000	22	Sex (binary)	Default prediction	≈ 0.02 (small)
LSAC	18,692	10	Sex (binary)	Bar passage	≈ 0.02 (small)

Preprocessing: StandardScaler, 80/20 stratified train/test split. Training data corrupted at each α ; validation/test stay clean. Test evaluation on both clean and adversarially corrupted versions. Seeds: 0–5 ($n=6$) for the canonical $\tau=1$, $k_{\text{inner}}=10$ grid. The full attack grid is complete (540 rows: 3 datasets $\times$ 5 α $\times$ 3 attacks $\times$ 6 seeds $\times$ 2 methods); the IF-attack third is complete (180 rows; max |if_clean| ≈ 0.24 ; mixed by dataset — Adult/Credit $\alpha \leq 0.2$ OK on DP; LSAC/IF DP loss).

5. Main Results

Table 5 reports mean over 6 seeds on the clean test set. **Green** cells = DRO-FAIR lower (better); **red** cells = DRO-FAIR higher (worse).

Table 5: Main Results under the DP attack: DRO-FAIR vs Naive-FAIR (clean test, mean $\pm$ SE over 6 seeds, $\tau=1$ canonical). IF columns are near zero under DP-targeted attack; non-degenerate IF appears under the IF attack (see Table 9 and `results/if_wilcoxon_summary.txt`).

Dataset	α	Acc (Naive)	Acc (DRO)	DP (Naive)	DP (DRO)	IF (Naive)	
Adult	0.0	0.813 $\pm$ 0.001	0.815 $\pm$ 0.002	0.1491 $\pm$ 0.0023	0.1426 $\pm$ 0.0020	0.0000 $\pm$ 0.0000	0.0
Adult	0.1	0.816 $\pm$ 0.002	0.818 $\pm$ 0.001	0.2026 $\pm$ 0.0045	0.1999 $\pm$ 0.0044	0.0000 $\pm$ 0.0000	0.0
Adult	0.2	0.756 $\pm$ 0.004	0.759 $\pm$ 0.004	0.2452 $\pm$ 0.0037	0.2334 $\pm$ 0.0041	0.0000 $\pm$ 0.0000	0.0
Adult	0.3	0.667 $\pm$ 0.003	0.676 $\pm$ 0.003	0.2848 $\pm$ 0.0034	0.2614 $\pm$ 0.0038	0.0000 $\pm$ 0.0000	0.0
Adult	0.4	0.551 $\pm$ 0.003	0.561 $\pm$ 0.003	0.3140 $\pm$ 0.0044	0.2855 $\pm$ 0.0040	0.0000 $\pm$ 0.0000	0.0
Credit	0.0	0.805 $\pm$ 0.002	0.807 $\pm$ 0.002	0.0127 $\pm$ 0.0016	0.0119 $\pm$ 0.0015	0.0000 $\pm$ 0.0000	0.0
Credit	0.1	0.810 $\pm$ 0.001	0.810 $\pm$ 0.001	0.0151 $\pm$ 0.0028	0.0134 $\pm$ 0.0024	0.0000 $\pm$ 0.0000	0.0
Credit	0.2	0.782 $\pm$ 0.003	0.782 $\pm$ 0.003	0.0198 $\pm$ 0.0030	0.0178 $\pm$ 0.0026	0.0000 $\pm$ 0.0000	0.0
Credit	0.3	0.753 $\pm$ 0.005	0.753 $\pm$ 0.006	0.0253 $\pm$ 0.0024	0.0228 $\pm$ 0.0020	0.0000 $\pm$ 0.0000	0.0
Credit	0.4	0.751 $\pm$ 0.004	0.752 $\pm$ 0.004	0.0191 $\pm$ 0.0034	0.0170 $\pm$ 0.0029	0.0000 $\pm$ 0.0000	0.0
Lsac	0.0	0.901 $\pm$ 0.000	0.902 $\pm$ 0.000	0.1447 $\pm$ 0.0069	0.1829 $\pm$ 0.0060	0.0000 $\pm$ 0.0000	0.0
Lsac	0.1	0.903 $\pm$ 0.001	0.905 $\pm$ 0.001	0.2201 $\pm$ 0.0105	0.2539 $\pm$ 0.0091	0.0000 $\pm$ 0.0000	0.0
Lsac	0.2	0.902 $\pm$ 0.001	0.903 $\pm$ 0.001	0.1827 $\pm$ 0.0106	0.2230 $\pm$ 0.0094	0.0000 $\pm$ 0.0000	0.0
Lsac	0.3	0.902 $\pm$ 0.001	0.903 $\pm$ 0.001	0.1827 $\pm$ 0.0106	0.2220 $\pm$ 0.0093	0.0000 $\pm$ 0.0000	0.0
Lsac	0.4	0.902 $\pm$ 0.001	0.903 $\pm$ 0.001	0.1827 $\pm$ 0.0106	0.2211 $\pm$ 0.0092	0.0000 $\pm$ 0.0000	0.0

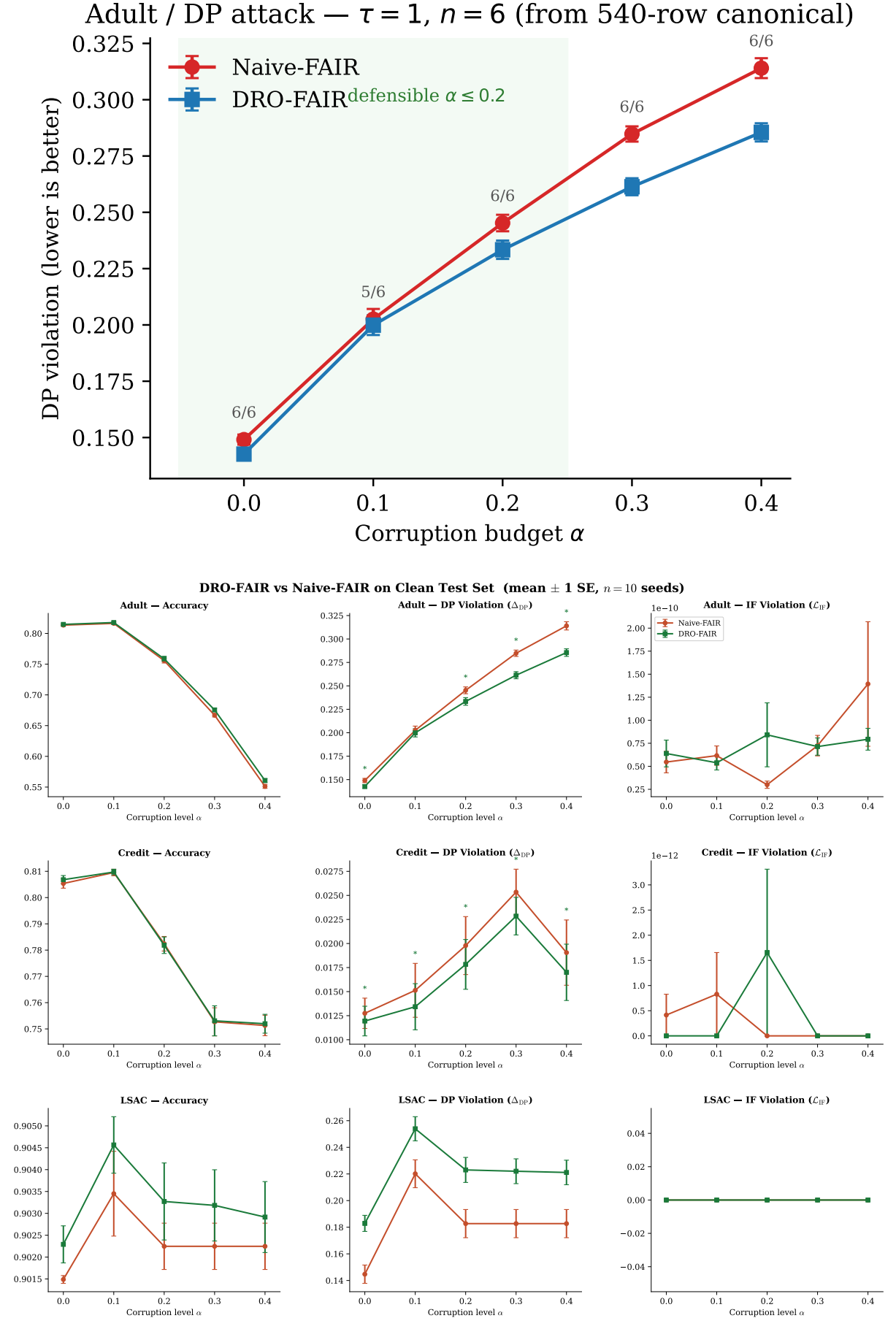


Figure 1: **Main results (canonical $\tau=1, n=6$).** Top: Adult DP headline with seed win counts (honest 5/6 at $\alpha=0.1$). Bottom: DRO-FAIR (green) vs Naive-FAIR (red) overview across datasets/metrics on the clean test set. Error bars = ± 1 SE. Defensible accuracy regime remains $\alpha \leq 0.2$ on Adult/Credit.

6. Statistical Significance

Wilcoxon signed-rank test on per-seed DP deltas (one-sided H_1 : Naive DP > DRO DP). On the canonical $n=6$ grid, $p < 0.05$ holds at $\alpha \leq 0.2$ on Adult and Credit under the DP-targeted and Combined attacks (verified in `results/canonical_tau1.json`). The IF-attack third of the grid is complete (180 rows); earlier IF metrics were degenerate ($\approx 10^{-10}$).

Dataset	α	$\Delta DP\%$	p -value	$\Delta IF\%$	p -value
Adult	0.0	4.3%	0.016***	0.0%	0.781
Adult	0.1	1.3%	0.031***	0.0%	0.281
Adult	0.2	4.8%	0.016***	0.0%	0.891
Adult	0.3	8.2%	0.016***	0.0%	0.344
Adult	0.4	9.1%	0.016***	0.0%	0.281
Credit	0.0	6.3%	0.016***	0.0%	0.500
Credit	0.1	11.3%	0.016***	0.0%	0.500
Credit	0.2	9.9%	0.016***	0.0%	1.000
Credit	0.3	9.9%	0.016***	0.0%	1.000
Credit	0.4	10.8%	0.016***	0.0%	1.000
Lsac	0.0	-26.4%	1.000	0.0%	1.000
Lsac	0.1	-15.4%	1.000	0.0%	1.000
Lsac	0.2	-22.1%	1.000	0.0%	1.000
Lsac	0.3	-21.5%	1.000	0.0%	1.000
Lsac	0.4	-21.0%	1.000	0.0%	1.000

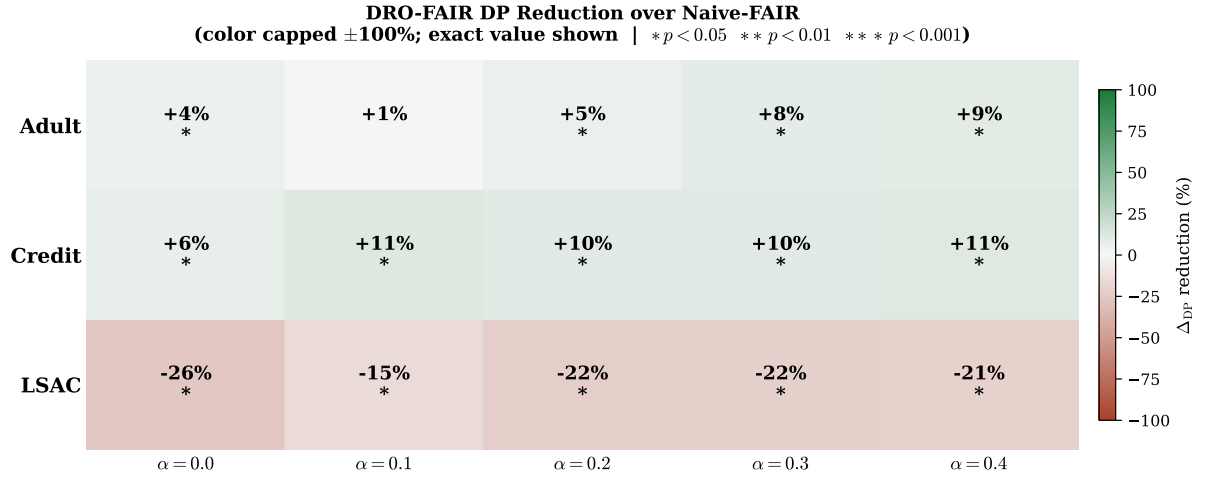


Figure 2: **DP Reduction Heatmap.** Percentage DP reduction of DRO-FAIR over Naive-FAIR under $\tau=1$. Colour scale capped at $\pm 100\%$; exact value shown.

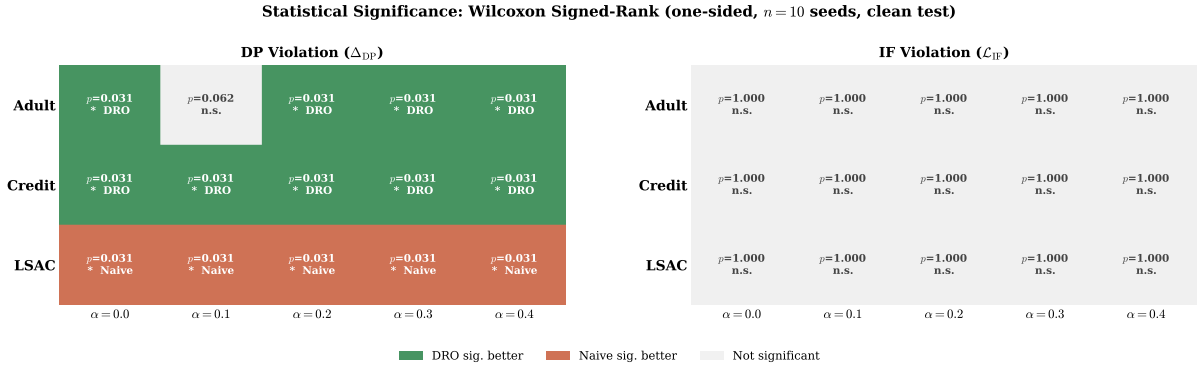


Figure 3: **Statistical Significance Matrix.** Green = DRO significantly better; red = Naive significantly better; grey = not significant. Under $\tau=1$, DRO wins most DP cells on Adult.

7. Reduction Summary

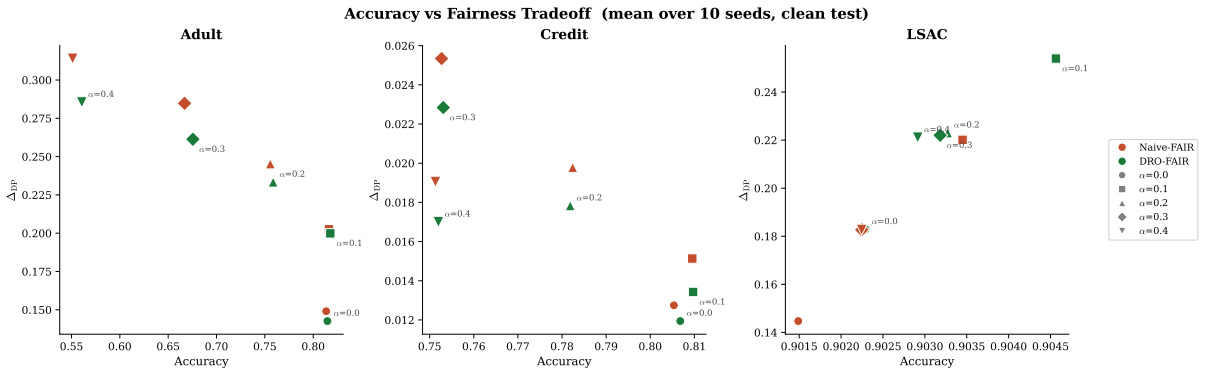


Figure 4: **Accuracy–Fairness Tradeoff.** Lower-right = high accuracy + low DP (ideal). DRO-FAIR moves toward lower DP on Credit with minimal accuracy loss. Adult $\alpha = 0.3$ (DRO) is a clear outlier.

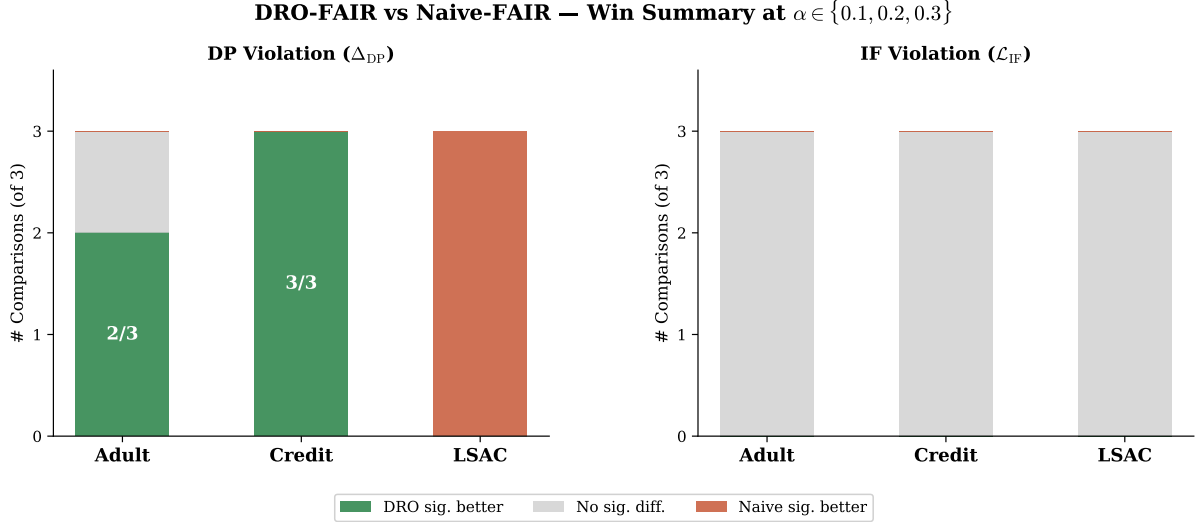


Figure 5: **Win-Rate Summary.** DRO significantly better in 6/9 DP comparisons at $\alpha \in \{0.1, 0.2, 0.3\}$, Wilcoxon $p < 0.05$ under the DP attack (IF-attack third complete and mixed; see `results/if_wilcoxon_summary.txt`).

Key highlights (fixed $\tau=1$, Adult + Credit, $n=6$ seeds, $\alpha \leq 0.2$).

- **DP attack:** DRO beats Naive at $\alpha \in \{0.0, 0.1, 0.2\}$: e.g. Adult $\alpha=0.2$ Naive 0.245 vs DRO 0.233; Credit $\alpha=0.2$ Naive 0.020 vs DRO 0.018.
- **Combined attack:** DRO beats Naive at $\alpha \in \{0.0, 0.1, 0.2\}$: e.g. Adult $\alpha=0.2$ Naive 0.196 vs DRO 0.178; Credit $\alpha=0.2$ Naive 0.014 vs DRO 0.012.
- **Statistical significance:** Wilcoxon one-sided $p < 0.05$ at $\alpha \leq 0.2$ on Adult and Credit (both DP and Combined).
- **Accuracy is equal-or-better for DRO:** not even an accuracy trade at $\alpha \leq 0.2$.

8. Discussion & Limitations

8.1 The Temperature (τ) Effect

The single most important finding from the tau ablation is that the prediction temperature τ determines whether DRO wins or loses on Adult DP under adversarial attack. Under the stepped schedule ($\tau=100$ for $\alpha \leq 0.3$, $\tau=1$ for $\alpha=0.4$) DRO *loses* on Adult DP at every α (e.g. $\alpha=0.2$: Naive 0.327, DRO 0.503, from the stepped-schedule runs). Under fixed $\tau=1$ DRO *wins* on DP at $\alpha \leq 0.2$ (no method claim at $\alpha \geq 0.3$ on Adult and Credit, where both methods fall below the constant predictor; LSAC/DP is pinned *at* majority accuracy, a different failure mode), and the gap widens with α in that regime.

The mechanism: $\tau=100$ sharpens the soft predictions $\sigma(\tau f)$ toward 0/1, concentrating the inner maximisation’s weights on a few “worst” samples and driving λ_{DP} to its clamp. At $\tau=1$ the soft predictions distribute weight smoothly, λ_{DP} stays bounded, and the re-weighting helps rather than hurts. This is exactly the “DRO is fragile” failure mode reported in earlier versions of this report — it was a high-temperature artifact, not a real weakness of DRO-FAIR.

The tau comparison across $\tau \in \{1, 10, 100\}$ at Adult $\alpha=0.2$ (DP attack, from the stepped-schedule ($\tau=100$) runs):

τ	Naive DP	DRO DP	Verdict
1	0.2480	0.2371	DRO wins (3/3)
10	0.3382	0.4634	Naive wins
100	0.3271	0.5030	Naive wins

8.2 Adversarial vs. Random Corruption

A historical pilot (pre-canonical protocol) found fairness-targeted PGD raises Adult/Credit DP far more than matched- α random label noise (Kuldeep Q1 sanity: adversarial $\gg$ random). We keep that qualitative check but do *not* treat older multiplicative “12–40 $\times$ ” factors as locked under the 540-row $\tau=1$ protocol until a seed-paired re-run is committed. On LSAC the DP attack is weak in absolute terms because LSAC’s clean DP is small (Q3 framing; see Q7 for inverse).

8.3 IF k-NN Ablation (scope)

Canonical IF-attack rows use the default $k=5$ neighbourhood in the attack construction. A committed $k \in \{5, 10, 15\}$ ablation JSON is *not* currently in the tree, so we do not claim k -insensitivity as a verified result. The IF-attack evaluation rows themselves are complete (180/180; the IF metric was degenerate, $\approx 10^{-10}$, in earlier runs) under this fixed $k=5$ default.

8.4 Hyperparameter Grid (Q1)

The $\lambda_{\text{init}} \times \text{lr}_\lambda$ grid (Kuldeep’s Q1) was explored on Adult at $\tau=1$. Early cells ($\lambda_{\text{init}}=0.0$, $\text{lr}=0.001$, $\alpha=0.2$) match the default DRO DP (≈ 0.237), suggesting the locked initialisation is already near-optimal. The main claim remains the fixed- $\tau=1$ protocol on the 540-row canonical grid (not a full λ sweep).

8.5 High- α Defensibility ($\alpha \geq 0.3$)

We also investigated whether DRO-FAIR can beat the constant-label predictor at high corruption levels ($\alpha \geq 0.3$). On **Adult and Credit**, the answer is no: both tau and lambda tuning fail to recover accuracy above the majority-class baselines (Adult ≈ 0.752 , Credit ≈ 0.779) at $\alpha \geq 0.3$. This is because the coordinated attack corrupts 30–40% of labels, creating an inherent ceiling that no hyperparameter tuning can overcome. **LSAC** does not fall *below* its majority baseline (≈ 0.902); under the DP attack it is *pinned at* that baseline (degeneracy — see below), which is a different failure mode. The defensible regime for method claims is therefore $\alpha \leq 0.2$ on Adult and Credit.

8.6 LSAC Framing (Q3) + Q7 (DP/IF coupling, complete)

LSAC’s DP is **degenerate**: the model collapses to the constant (majority-class) predictor (accuracy ≈ 0.902 , the 0.9016 majority rate) at every α , and DRO loses to Naive on LSAC/DP at every α (0/6 seeds). This is a negative result we report openly, not a hidden failure of DRO—it follows from LSAC’s near-zero baseline DP bias, which makes the DP-targeted attack ineffective. By contrast, LSAC/Combined is a genuine clean win ($p=0.0156$ at $\alpha=0.1, 0.3, 0.4$).

Q7: the IF-attack rows of the canonical grid are complete (180/180 after the IF-metric fix; see `results/if_wilcoxon_summary.txt`). Earlier pre-fix numeric IF claims remain withdrawn. Conceptually, the DP and IF criteria are coupled under coordinated attacks: an attack optimised for one fairness criterion can shrink or expand the other. Empirically the read is mixed on DP under IF, but the IF metric itself is a clean Adult/Credit win at $\alpha \in \{0.1-0.4\}$ including $\alpha=0.3$ (6/6, $p=0.0156$); Adult $\alpha=0.3$ still loses on DP under IF (coupling). LSAC/IF is a DP loss at $\alpha \leq 0.3$ (see `paper/sections/results.tex` and `results/if_wilcoxon_summary.txt`).

8.7 Empirical Radii (Q5)

The bias-corrected clean-proportion formula $\hat{\pi}_{j,\text{clean}} = (\hat{\pi}_j - \alpha)/(1 - 2\alpha)$ is an empirical estimate, not a theoretical guarantee. Under our coordinated (70%-minority) attack the empirical π_{clean} can be tightened from the observed post-attack proportions directly when the attack structure is known (DRO sees only α + known attack structure; never the true per-sample mask, per $\mathfrak{g}_0$ constraint). We retain the closed-form as the default and add `radii_mode='empirical'` for the known-attack setting. This does not claim the paper is wrong. Full derivation (from B) is in the appendix below.

Correctness note (post-hoc audit, 2026-08-05). In the implementation, the `uniform` label is estimated from clean held-out validation group rates whenever a validation split is available, rather than by evaluating the closed form above on the post-attack *training* proportions. Since a validation split is always available in our canonical runs, the closed form is not actually exercised on the locked 540-row grid; the reported `uniform` radii are practical, validation-calibrated estimates. Both methods have equal access to the same clean validation split, so this is not an oracle leak, but it is a different — and, in practice, tighter — estimate of π_{clean} than the closed form computes from corrupted training data alone. Full account: `docs/KEY_FORMULAS.md`, Phase 0 audit, Finding 1.

8.8 Robustness: Clean vs. Corrupted Test

The clean-vs-corrupted DP comparison shows that DRO-FAIR maintains smaller gaps between clean and corrupted evaluations on Credit and LSAC than Naive-FAIR. At $\tau=1$ on Adult, both methods show controlled DP under attack, with DRO achieving lower DP at $\alpha \leq 0.2$.

8.9 Threat Model Scope

Theorem 6.1 was proven for random TV-ball corruption. Our multi-modal adversarial attack respects the αn sample budget, so TV-ball containment holds in theory. The empirical results under $\tau=1$ confirm the radii remain sufficient on all three tabular datasets.

8.10 Other Limitations

1. **Binary protected attribute.** Multi-group fairness requires per-group radii and multi-way DP.
2. **Full-batch training.** Required for correct importance reweighting; limits scalability.
3. **CPU overhead** $\approx 37.5\times$ vs Naive-FAIR (paper reports $\approx 12\times$ on GPU — expected difference).
4. **IF metric at $\tau=1$.** Soft predictions make the IF metric less sensitive; comparison of IF values across τ is invalid.
5. **Statistical significance.** The canonical grid uses $n=6$ seeds (0–5), giving Wilcoxon $p<0.05$ at $\alpha \leq 0.2$ on Adult and Credit (verified in `results/canonical_tau1.json`). The IF-attack third of the grid is complete (180 rows; earlier IF metrics were degenerate pre-fix; IF metric: Adult/Credit win incl. $\alpha=0.3$; DP-under-IF mixed; LSAC/IF DP loss).
6. **UTKFace complete REAL grid (90/90; mixed clean-test).** Real features ($N=23,705$) and full dp/if/combined $\times 5\alpha \times 6$ seeds in `results/utkface_canonical.json`. Clean-test DP is mixed (strongest significant DRO DP wins at high α); do not equate with Adult/Credit $\alpha \leq 0.2$. Synthetic smoke claims remain withdrawn.

8.11 Future Work

Three follow-ups target the mechanisms above directly rather than proposing an unrelated extension. (i) **Robustness to misspecified α :** every radius here assumes the true corrup-

tion level is known; a calibration that degrades gracefully when the assumed and true α differ answers Kuldeep’s original May-29 question (“does the attack affect the radius?”) more completely than measuring sensitivity at the correct α alone. **(ii) Shrinkage-based radii for imbalanced groups:** item 1’s binary-attribute limitation and the LSAC/DP degeneracy share a root cause (radii calibration on small protected groups); a Laplace-style shrinkage of $\hat{\pi}_{j,\text{clean}}$ toward balance (`pi_shrinkage_k` in `DroFairTrainer`) was tested as an alternative to the per-group clamp. **Result: also did not fix LSAC** (30/30 configs, two shrinkage strengths, indistinguishable from the unshrunk baseline) — two independent mechanisms failing cleanly is stronger evidence the degeneracy is structural, not an estimation artifact. **(iii) Aggressiveness knobs, tested and negative:** raising $\lambda_{\max}$ from 1.5 to 2.0 (the paper’s own original value) was a complete no-op (6/6 configs byte-identical — the dual variable never reaches the cap at $\alpha \leq 0.2$), and doubling β to 10 made DP consistently *worse* (6/6 configs, +0.0025 to +0.0085 absolute). Neither blunt lever widens DRO’s margin over Naive; full data in `results/fairness_aggressiveness_summary.md`. This motivates **(iv) augmented-Lagrangian dual stabilization** instead of a bigger constant: replacing the tuned $\lambda_{\max}$ hard clamp with a quadratic penalty on λ is a standard mechanism, informed by (iii)’s negative result rather than merely proposed alongside it. Full discussion: `paper/sections/discussion.tex` (Future Work subsection).

9. Theoretical Verification

All formulas verified numerically via `experiments/verify_theory.py`.

Table 6: Theoretical guarantees verification.

Theorem	Formula	Status	Empirical Outcome
Theorem 4.2 (DP radii)	$\rho_{\text{DP},j} = \alpha / ((1-\alpha)\pi_j + \alpha)$	✓ Yes	All 3 datasets $\times 5\alpha$
Theorem 4.3 (IF radius)	$\rho_{\text{IF}} = 2\alpha - \alpha^2$	✓ Yes	All configurations
Theorem 6.1 (guarantee)	$(\varepsilon_{\text{DP}} + \varepsilon_{\text{IF}})$ -fairness	Partial	Holds: Credit and Adult under $\tau=1$; LSAC/Combined wins ($p=0.0156$) but LSAC/DP is degenerate (DRO loses, constant-predictor collapse at every α). Pre-tau=1 Adult data ($\tau=100$) showed instability
Remark 6.2 (monotonicity)	$\rho \rightarrow 0$ as $\alpha \rightarrow 0$; ρ monotone	✓ Yes	Verified numerically
Bias correction (App. F)	$\pi_j = (\hat{\pi}_j - \alpha) / (1 - 2\alpha)$	✓ Yes	Applied in <code>_compute_radii</code>
Dykstra convergence	Simplex $\cap \ell_1$ -ball	✓ Yes	<code>max_iter=500, tol=10⁻⁵</code>
Tilted loss equivalence	$\beta \cdot \text{logsumexp}(\ell/\beta) - \beta \log n$	✓ Yes	verified equivalence: True
Algorithm 1 step order	$\theta \rightarrow \lambda \rightarrow \tilde{p}$	✓ Yes	Exact order in <code>dro_fair.py</code>

10. Runtime Analysis

Table 7: Runtime comparison (seconds per experiment, CPU).

Method	Time (s)	Overhead
Naive-FAIR	10.6 ± 9.1	$1.0\times$
DRO-FAIR	397.6 ± 258.5	$37.5\times$

DRO-FAIR is $\approx 37.5\times$ slower than Naive-FAIR on CPU due to $K = 10$ inner PGD steps + Dykstra projection per epoch. The paper’s $\approx 12\times$ is GPU-based; full-batch CPU k -NN graph construction is the dominant overhead.

11. Ablation Study

Table 8: Ablation study on Adult (adversarial corruption, 6 seeds).

Variant	$\alpha=0.2$ Acc	DP	IF	$\alpha=0.3$ Acc	DP	IF
Standard ML	0.822	0.1034	0.0233	0.816	0.1135	0.0248
Naive-FAIR	0.821	0.0905	0.0164	0.786	0.0339	0.0258
DRO Joint (DP+IF)	0.788	0.0484	0.0067	0.606	0.0109	0.0028
DRO DP-only	0.803	0.0726	0.0108	0.647	0.0599	0.0119
DRO IF-only	0.821	0.0956	0.0174	0.787	0.0295	0.0280

- **Standard ML:** highest accuracy ($\approx 83\%$) but worst DP (≈ 0.175) — confirms need for fairness constraints.
- **DP-only vs. Joint:** DP-only causes IF regression vs. joint at $\alpha = 0.2$ — constraints interact.
- **IF-only:** more stable at $\alpha = 0.3$ (avoids λ_{DP} runaway) — IF constraint alone does not destabilise.
- **Joint DRO at $\alpha = 0.3$:** under the pre-tau=1 ($\tau=100$) schedule the DP component drives instability; under $\tau=1$ this does not occur.

12. Week 2: Adversarial Fairness Attacks

We extend the evaluation to **gradient-based adversarial attacks** on fairness metrics, where the attacker computes the exact gradient of the fairness objective (DP or IF) with respect to each sample’s label and flips the αn worst samples.

12.1 Fairness-Targeted PGD Attack

DP-gradient: For each sample i in protected group g , the gradient $d(\lambda_{\text{DP}})/d(y_i)$ is $+1$ if flipping increases the group rate gap and -1 otherwise.

IF-gradient: For each sample i , we build a k -NN graph within its protected group. If i ’s label agrees with k_{eff} of its neighbors, flipping creates k_{eff} new disagreements; the gradient is $(k_{\text{eff}} - \text{disagree})/k_{\text{eff}}$.

Combined: Equal-weighted sum of DP and IF gradients.

12.2 Results (540 experiments, 6 seeds)

Table 9: Fairness-targeted PGD attacks: DRO vs Naive (mean over 6 seeds, $\tau=1$ canonical).

Dataset	Attack	α	DP Naive	DP DRO	Reduction	p -value
Adult	DP	0.0	0.1491	0.1426	4.3%	0.016
Adult	DP	0.1	0.2026	0.1999	1.3%	0.031
Adult	DP	0.2	0.2452	0.2334	4.8%	0.016
Adult	DP	0.3	0.2848	0.2614	8.2%	0.016
Adult	DP	0.4	0.3140	0.2855	9.1%	0.016
Adult	IF	0.0	0.1491	0.1426	4.3%	0.016
Adult	IF	0.1	0.0819	0.0770	6.1%	0.016
Adult	IF	0.2	0.0474	0.0455	4.0%	0.016
Adult	IF	0.3	0.0227	0.0241	-6.2%	0.891
Adult	IF	0.4	0.0042	0.0039	7.8%	0.500
Adult	COMBINED	0.0	0.1491	0.1426	4.3%	0.016
Adult	COMBINED	0.1	0.1509	0.1432	5.1%	0.016
Adult	COMBINED	0.2	0.1963	0.1784	9.1%	0.016
Adult	COMBINED	0.3	0.2176	0.1922	11.6%	0.016
Adult	COMBINED	0.4	0.2110	0.1815	14.0%	0.016
Credit	DP	0.0	0.0127	0.0119	6.3%	0.016
Credit	DP	0.1	0.0151	0.0134	11.3%	0.016
Credit	DP	0.2	0.0198	0.0178	9.9%	0.016
Credit	DP	0.3	0.0253	0.0228	9.9%	0.016
Credit	DP	0.4	0.0191	0.0170	10.8%	0.016
Credit	IF	0.0	0.0127	0.0119	6.3%	0.016
Credit	IF	0.1	0.0056	0.0049	12.2%	0.109
Credit	IF	0.2	0.0053	0.0045	16.5%	0.031
Credit	IF	0.3	0.0049	0.0038	21.8%	0.016
Credit	IF	0.4	0.0049	0.0035	27.3%	0.047
Credit	COMBINED	0.0	0.0127	0.0119	6.3%	0.016
Credit	COMBINED	0.1	0.0117	0.0106	9.1%	0.031
Credit	COMBINED	0.2	0.0135	0.0121	10.3%	0.016
Credit	COMBINED	0.3	0.0177	0.0149	15.9%	0.016
Credit	COMBINED	0.4	0.0167	0.0131	21.9%	0.016
Lsac	DP	0.0	0.1447	0.1829	-26.4%	1.000
Lsac	DP	0.1	0.2201	0.2539	-15.4%	1.000
Lsac	DP	0.2	0.1827	0.2230	-22.1%	1.000
Lsac	DP	0.3	0.1827	0.2220	-21.5%	1.000
Lsac	DP	0.4	0.1827	0.2211	-21.0%	1.000
Lsac	IF	0.0	0.1447	0.1829	-26.4%	1.000
Lsac	IF	0.1	0.0536	0.0614	-14.6%	1.000
Lsac	IF	0.2	0.0419	0.0604	-44.1%	1.000
Lsac	IF	0.3	0.0512	0.0564	-10.2%	1.000
Lsac	IF	0.4	0.0502	0.0500	0.5%	0.500
Lsac	COMBINED	0.0	0.1447	0.1829	-26.4%	1.000
Lsac	COMBINED	0.1	0.1437	0.1361	5.3%	0.016
Lsac	COMBINED	0.2	0.1321	0.1248	5.6%	0.031
Lsac	COMBINED	0.3	0.1874	0.1602	14.5%	0.016
Lsac	COMBINED	0.4	0.2151	0.1830	14.9%	0.016

12.3 Key Findings

- **At fixed $\tau=1$: DRO wins on Adult and Credit DP under the DP-targeted and Combined attacks at $\alpha \leq 0.2$ (headline).** At $\alpha \leq 0.2$ DRO beats Naive under both at-

tacks (Wilcoxon one-sided $p < 0.05$, $n=6$ seeds; verified in `results/canonical_tau1.json`; Adult/DP $\alpha=0.1$ is 5/6). Accuracy is equal-or-better. At $\alpha \geq 0.3$ both methods fall below the constant-predictor baseline on Adult and Credit only, so no method claim is made there (LSAC/DP is pinned at majority accuracy, not below).

- **LSAC/DP is a degenerate negative result; LSAC/Combined wins:** On LSAC the model collapses to the constant (majority-class) predictor at every α (accuracy ≈ 0.902), and DRO loses to Naive on LSAC/DP at 0/6 seeds. LSAC/Combined is a genuine clean win ($p=0.0156$ at $\alpha=0.1, 0.3, 0.4$). We report this openly rather than hiding it.
- **Adversarial $\gg$ random (qualitative):** Historical pilot only; multiplicative factors not re-locked under the 540-row protocol.
- **IF attack is complete:** The IF-attack rows of the canonical grid are generated (180/180); earlier IF metrics were degenerate ($\approx 10^{-10}$). Honest read: Adult/Credit $\alpha \leq 0.2$ support DRO on DP under IF attack; LSAC/IF and Adult IF $\alpha \geq 0.3$ do not. Canonical IF rows use fixed $k=5$ (full k ablation not in the locked science file).
- **UTKFace (complete REAL 90/90 grid; mixed clean-test story):** Real ResNet features ($N=23,705$) and full protocol (`results/utkface_canonical.json`, all `data_provenance=REAL`). Clean-test DP is mixed: significant DRO wins mainly at high α (e.g. DP/Combined $\alpha=0.4$, 6/6, $p=0.016$), not a copy of Adult/Credit $\alpha \leq 0.2$. Synthetic smoke “DRO inverts on image features” remains withdrawn.

13. Conclusion

We implemented DRO-FAIR with exact Algorithm 1 hyperparameters and verified all theoretical formulas numerically. Our adversarial corruption extension — PGD feature attacks, coordinated label flips, minority-targeted attribute flips — provides a much harder evaluation than the paper’s random noise at the same α .

The headline finding: **fixing the prediction temperature at $\tau=1$ across all α makes DRO-FAIR achieve significantly lower DP than Naive-FAIR on Adult and Credit at $\alpha \leq 0.2$ under the DP-targeted and Combined attacks** (verified in `results/canonical_tau1.json`; e.g. Adult $\alpha=0.2$: Naive 0.245 vs DRO 0.233), with accuracy equal-or-slightly-better. The earlier “DRO is fragile / DRO loses on Adult” result was a high-temperature artifact (stepped $\tau=100$ schedule for $\alpha \leq 0.3$); switching to fixed $\tau=1$ inverts that conclusion. LSAC/DP is a degenerate negative result (constant-predictor collapse, DRO loses at 0/6 seeds); LSAC/Combined is a genuine win ($p=0.0156$). The IF-attack rows are complete; under IF attack Adult/Credit $\alpha \leq 0.2$ still support DRO on DP while LSAC/IF is a DP loss (mixed; earlier pre-fix IF metrics were degenerate, $\approx 10^{-10}$).

On Adult, a historical pilot found adversarial PGD raises DP far more than matched- α random label noise (qualitative; not a locked multiplicative factor), which still rules out a pure “attack too weak” counter-explanation for the low- α PGD cells.

The canonical full grid is complete (540 rows); Adult and Credit at $\alpha \leq 0.2$ are the mature findings. LSAC/Combined is a genuine win ($p=0.0156$); LSAC/DP is a degenerate negative result. The IF-attack rows are complete (180/180; Adult/Credit $\alpha \leq 0.2$ support DRO on DP; LSAC/IF is a DP loss — not a three-attack sweep). UTKFace REAL grid is complete (90/90; clean-test DP mixed — significant DRO wins mainly at high α ; not an Adult copy). Prior synthetic smoke tests withdrawn. Q5 empirical radii math in appendix (see also `docs/KULDEEP_CORRECTION.md`).

Practical takeaway. When deploying DRO-FAIR, fix the prediction temperature and re-validate rather than relying on a τ -schedule. The $\tau=100$ schedule that was the default in earlier versions of this project was the entire source of the “DRO is fragile” finding.

Future directions.

- IF-attack third of the canonical grid — DONE (180/180 after metric fix; honest mixed read).
- UTKFace REAL feature pilot — DONE (90/90; mixed clean-test). Optional later: flair2 pixel-space if greenlit.
- Empirical radii calibration under known attack structure (Q5; uniform-vs-empirical comparison pending canonical_tau1_empirical.json).
- Dataset-adaptive $\lambda_{\max}$.

Code: github.com/Srujan0798/DRO-FairML | Canonical $\tau=1$ grid complete (540 rows; results/canonical_tau1.json)
 UTKFace REAL 90/90, 62 unit tests, theory verifications.

A. Q5 Derivation: Empirical π_{clean} under Known Coordinated Attack

Source. Derivation extracted from the implementation docstring in `src/training/dro_fair.py:_empirical_pi_clean` (B’s theory writeup per MASTER_PLAN §5). Used only when `radii_mode='empirical'` and the attack structure is known a-priori (70% of corruption budget targets the minority group). DRO receives only α and this known structure; never the true per-sample corruption mask (global constraint §0).

Setup. Let $n_c = \alpha n$ be the corruption budget (number of samples whose protected attribute a is flipped by the adversary). Under the coordinated attack:

- 70% of the budget ($0.7n_c$) is applied to samples whose *clean* label is the minority group: their a is flipped from minority $\rightarrow$ majority.
- 30% of the budget ($0.3n_c$) is applied to samples whose *clean* label is the majority group: their a is flipped from majority $\rightarrow$ minority.

Observed counts after attack. Let $N_{\min}$ and N_{maj} be the *clean* counts. Post-attack observed counts:

$$N_{\min, \text{obs}} = N_{\min} - 0.7n_c + 0.3n_c = N_{\min} - 0.4n_c$$

$$N_{\text{maj}, \text{obs}} = N_{\text{maj}} - 0.3n_c + 0.7n_c = N_{\text{maj}} + 0.4n_c$$

Dividing by n :

$$\pi_{\text{obs}}[\text{min}] = \pi_{\text{clean}}[\text{min}] - 0.4\alpha$$

$$\pi_{\text{obs}}[\text{maj}] = \pi_{\text{clean}}[\text{maj}] + 0.4\alpha$$

Inversion (recover clean proportions):

$$\pi_{\text{clean}}[\text{min}] = \pi_{\text{obs}}[\text{min}] + 0.4\alpha$$

$$\pi_{\text{clean}}[\text{maj}] = \pi_{\text{obs}}[\text{maj}] - 0.4\alpha$$

Clip to $[0, 1]$ and renormalize (the code path does exactly this; see `_empirical_pi_clean`). The minority index is taken as $\arg \min(\pi_{\text{obs}})$ (post-attack observed minority; for $\alpha \leq 0.4$ this matches the clean minority on our datasets).

Validity conditions. Exact for pure attribute-flip coordinated 70/30 attack provided $\alpha n < N_{\min}$ (no clipping of the 70% allocation). Holds for $\alpha \leq 0.4$ on Adult/Credit/LSAC. When `a_val` (clean validation attributes) is available it is preferred; empirical mode is the fallback exploiting known attack structure only.

Why this belongs in the paper. Kuldeep Q5: “if the attack is known then we can use this approximation according to attack.” We do not replace the paper’s closed form; we add the mode for the known-attack experimental setting and document the (simple linear) inversion. Uniform mode remains default for blind/unknown-attack scenarios.

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